

Project 2 - Computer Vision 2025-2026

Object Detection for Waste Recognition

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1. Introduction

This project investigates the performance of modern object detectors for waste detection under different visual conditions. Three pretrained models, YOLO, DETR and RT-DETR, are first evaluated in a zero-shot setting on PASCAL VOC and the TACO waste dataset. A small RT-DETR model is then fine-tuned on TACO to enable category-level recognition. Finally, all detectors are tested on UAV-Vaste drone imagery to study how well they generalize to aerial viewpoints. The aim is to compare their detection behaviour across domains and identify key challenges for reliable waste detection in real scenarios.

2. Methodology

2.1. Task 1: Zero-shot evaluation on VOC

For Task 1 the three detectors (YOLO, DETR, RT-DETR) were evaluated in a zero-shot setting on a fixed VOC subset. Images were preprocessed according to each model's expected input format and all predictions were converted to a unified COCO-style representation (xyxy boxes, scores, integer labels). Ground-truth annotations were mapped to the same format.

Each model was evaluated at the confidence thresholds $\{0.10, 0.25, 0.50, 0.75, 0.90\}$. For every threshold, the `MeanAveragePrecision` metric from `TorchMetrics` was used to compute all required measures (mAP, mAP@50/75, MAR@1/10/100, and scale-based mAP). Results were stored in a single table indexed by model and threshold, allowing comparison of performance trends and threshold sensitivity.

2.2. Task 2: Zero-shot evaluation on TACO

The procedure of Task 1 was reused for the COCO-formatted TACO dataset. Images and annotations were loaded through `pycocotools`, and per-split category statistics were computed. All detectors were evaluated on the test set using two required selection strategies:

- **Top- k :** k equals the number of ground-truth boxes in the

image. Predictions were sorted by confidence and the top- k were retained.

- **IoU-top10:** the ten highest-confidence predictions were considered and each ground-truth box was matched to the prediction with the highest IoU, enforcing one-to-one matching.

Both strategies were applied across the same confidence thresholds used in Task 1. Metrics were computed using the same `TorchMetrics` pipeline to ensure full comparability. All results were collected into a unified table to analyse model stability, threshold sensitivity, and differences between top- k and IoU-based matching.

Finally, an additional comparison was performed between YOLO-11m and YOLO-11s at the best-performing confidence value, to assess the effect of model size in the zero-shot TACO scenario.

Additionally for Task 2.3, RT-DETR was fine-tuned on TACO using the same preprocessing pipeline and evaluation format adopted for the zero-shot experiments. The training split was augmented with colour and illumination variations, as well as Gaussian blur and random noise to encourage robustness to low-resolution details and sensor noise. The model was trained for 20 epochs and the best checkpoint was selected automatically using validation mAP.

Two equivalent logging configurations were supported in the training code: a `Weights&Biases` run for online tracking of losses and metrics, and a local-only mode where the trainer outputs the same metrics without external logging. In both cases, the best checkpoint was finally evaluated on the test split using the same post-processing and metric computation as before.

2.3. Task 3: Generalizing to Aerial/Drone Images

For Task 3 the focus shifted from ground-level images to aerial imagery. A subset of UAV-Vaste was selected, and ten representative frames were manually chosen to cover diverse conditions: different surface types such as grass, pavement, asphalt and vegetation, varying camera-to-ground distances, and scenes containing either sparse or

densely clustered waste. Task 3 was designed as a qualitative study.

All three detectors (YOLO, DETR, RT-DETR) were evaluated on the selected images using two confidence thresholds chosen directly from the quantitative findings of Tasks 1 and 2: a low threshold of 0.10, which yielded the highest mAP for YOLO (even higher than the other models at that value), and a higher threshold of 0.50, which is the largest threshold at which DETR and RT-DETR still achieve their maximum mAP before it starts to decrease. Images were processed in the same unified pipeline as before, and all detections were stored in COCO format.

Moreover, the fine-tuned RT-DETR from Task 2 was re-trained from its best checkpoint and run on the same UAV-Vaste images and confidence thresholds (0.10 and 0.50) as the other detectors. Inference and visualisation followed exactly the same pipeline.

3. Evaluation and Discussion

3.1. Task 1: VOC zero-shot results

Across all confidence thresholds, RT-DETR is the strongest and most stable model, achieving the highest mAP and MAR values (see Appendix Fig. 1). YOLO performs competitively only at the lowest threshold (0.10) and then degrades rapidly as confidence increases, while DETR remains more stable but consistently below RT-DETR. This behaviour reflects the typical precision–recall trade-off: increasing the threshold reduces the number of detections per image and therefore lowers recall. Scale-wise, all models struggle with small objects, whereas RT-DETR retains a clear advantage on medium and large categories (Appendix Fig. 3).

3.2. Task 2: TACO zero-shot results

The overall ranking is consistent with Task 1: RT-DETR performs best, followed by DETR, then YOLO. However, threshold sensitivity is more pronounced on TACO. YOLO briefly leads at 0.10, reaches DETR and RT-DETR at 0.25, and then its mAP drops sharply, deteriorating faster than in the VOC experiment. DETR and RT-DETR remain stable up to 0.50 and then decrease gradually, with RT-DETR preserving a small advantage throughout. All models obtain zero mAP on small objects, confirming the difficulty of detecting very small debris in a zero-shot setting.

Top- k vs IoU-top10. Both selection strategies produce similar trends and do not change the ranking of the models (see Appendix Fig. 5 and Fig. 6). IoU-top10 provides only marginal improvements by removing redundant detections, but the difference relative to top- k is negligible in practice.

YOLO-11m vs YOLO-11s. An additional comparison was carried out between the medium (YOLO-11m) and small (YOLO-11s) variants at the best-performing confidence threshold. YOLO-11m consistently achieved higher mAP and MAR values, indicating that the larger backbone transfers better to the TACO domain. YOLO-11s produced fewer correct detections and showed weaker localisation, although the overall behaviour of the two variants remained similar. This confirms that model capacity has a noticeable impact on zero-shot generalisation, even when the ranking relative to DETR and RT-DETR remains unchanged.

Fine-tuned RT-DETR The fine-tuned RT-DETR model shows a clear learning trend in the first epochs, with both training and validation losses decreasing steadily and closely together (see Appendix Fig. 7). This indicates that the model is effectively fitting the data without exhibiting overfitting, which would manifest as diverging curves. After roughly the midpoint of the schedule, loss values begin to oscillate and improvements slow down, suggesting that 20 epochs are sufficient for convergence under the chosen configuration.

Validation mAP consistently increases during training and reaches its maximum around epoch 12 (Appendix Fig. 8), supporting the checkpoint selection strategy based on best validation performance. The final performance saturates around mAP ≈ 0.27 , which is expected given the small backbone capacity and the limited size and variability of TACO.

Scale-specific analysis confirms the known behaviour of RT-DETR on this task: large objects are learned well, medium objects adequately, while small items remain the most challenging category. The fixed input size of 480×480 further reduces resolution on tiny instances and limits learning of fine details.

Class-wise metrics also exhibit high variability. This is strongly correlated with the long-tailed distribution of TACO categories: objects such as *Can* yield higher mAP due to larger size and more distinctive structure, while *Cigarette* remains difficult because of small size, thin shape and frequent background blending. Interestingly, *Plastic bag & wrapper* is very frequent in the dataset but not the best learned category, highlighting that appearance variability plays a stronger role than raw sample count.

Recall-based measures follow a slightly different pattern. MAR shows a declining trend even when precision improves, indicating that the model becomes more selective as it learns and occasionally fails to detect some ground-truth instances. This behaviour reflects dataset imbalance and the limited learning signal for rare or small classes.

Overall, the fine-tuned RT-DETR substantially outperforms the zero-shot configurations on the same data splits, confirming that domain adaptation is necessary for waste

detection. However, its generalisation is still constrained by resolution, backbone capacity and dataset characteristics.

3.3. Task 3: Generalizing to Aerial/Drone Images

Task 3 shows a clear drop in performance compared to Tasks 1 and 2. This is expected, since in UAV-Vaste the objects are much smaller due to the greater camera distance (see Appendix Fig. 9 and Fig. 10). Many items occupy only a few pixels, making detection considerably harder than in TACO.

Based on the results of Tasks 1 and 2, the expectation was that DETR and RT-DETR would still perform reliably at a confidence of 0.50, while YOLO would degrade quickly. This was only partially confirmed. At 0.10, DETR and RT-DETR detect more objects than YOLO but produce some false positives, whereas YOLO detects very few items and sometimes none, although almost never producing false positives.

When increasing the threshold to 0.50, YOLO collapses almost entirely, but DETR and RT-DETR also struggle more than anticipated under aerial conditions (Appendix Fig. 10). They reduce false positives, yet all models face difficulty detecting waste reliably when the confidence requirement becomes stricter.

Overall, RT-DETR remains the most effective model in this scenario: it retrieves the highest number of relevant objects while maintaining a manageable amount of false positives. YOLO, on the other hand, is clearly the weakest model, missing most objects and degrading rapidly as the threshold increases.

Fine-tuned RT-DETR. The fine-tuned model performs worse than the pretrained ones at both thresholds (Appendix Fig. 9). Its training did not include such extreme scale reductions, and although the aerial viewpoint is more regular, objects become too small to be recognised. As a result, the fine-tuned model struggles to generalise to UAV-Vaste and fails almost entirely under stricter confidence settings.

Conclusion

This project compared the behaviour of YOLO, DETR and RT-DETR across increasingly challenging scenarios for waste detection. Zero-shot experiments on VOC and TACO showed that RT-DETR is the most effective and consistent model overall in our setting, while YOLO performs competitively only at very low confidence thresholds. Fine-tuning RT-DETR on TACO improves performance on ground-level data, but this gain does not transfer to UAV imagery, where objects become extremely small and difficult to detect. These findings indicate that robust waste detection in real environments requires both scale-aware modelling and training data that better reflect deployment viewpoints.

The differences we observed can be linked to the architectural principles of the models. DETR uses a Transformer encoder with global self-attention and bipartite matching to produce a fixed number of predictions without requiring non-maximum suppression, which allows it to reason jointly over objects and improves localisation accuracy [1]. However, the original study also reports that DETR struggles specifically with small objects, performing best on medium and large ones [1], which aligns with our results on TACO and UAV-Vaste. RT-DETR builds on this design by introducing an efficient hybrid multi-scale encoder and a query selection mechanism that prioritises high-quality features, leading to improvements in both accuracy and real-time performance on COCO benchmarks [2]. In our experiments, these design choices translated into stronger detection and lower susceptibility to threshold changes compared to YOLO.

By contrast, YOLO relies on dense grid-based predictions and confidence scoring that make it efficient and accurate on objects with clearly visible features, but in our tests its recall dropped sharply when confidence increased, resulting in frequent missed detections on small instances. The fine-tuned RT-DETR model reduced the semantic gap to the ground-level domain but did not overcome the resolution limitations present in DETR-style approaches, and therefore struggled in aerial scenes dominated by tiny objects.

Overall, the behaviour of all models confirms that aerial waste detection is primarily limited by object scale. Addressing this challenge will likely require either higher-resolution feature extraction, domain adaptation targeted at drone viewpoints, or architectures explicitly designed to handle tiny objects.

References

- [1] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European Conference on Computer Vision (ECCV)*, pages 213–229. Springer, 2020. 3
- [2] Yian Zhao, Wenyu Lv, Shangliang Xu, Jinman Wei, Guanzhong Wang, Qingqing Dang, Yi Liu, and Jie Chen. DETRs beat YOLOs on real-time object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16954–16964, 2024. 3

Appendix

conf	0.10	0.25	0.50	0.75	0.90
model					
DETR	0.418	0.418	0.418	0.412	0.402
RT-DETR	0.621	0.621	0.621	0.569	0.483
YOLO	0.464	0.452	0.426	0.353	0.196

Figure 1. Task 1 — mAP vs confidence.

conf	0.10	0.25	0.50	0.75	0.90
model					
DETR	8.12	8.12	8.12	5.86	4.09
RT-DETR	4.16	4.16	4.16	2.79	1.49
YOLO	6.50	4.32	3.07	1.96	0.73

Figure 2. Task 1 — Average detections per image.

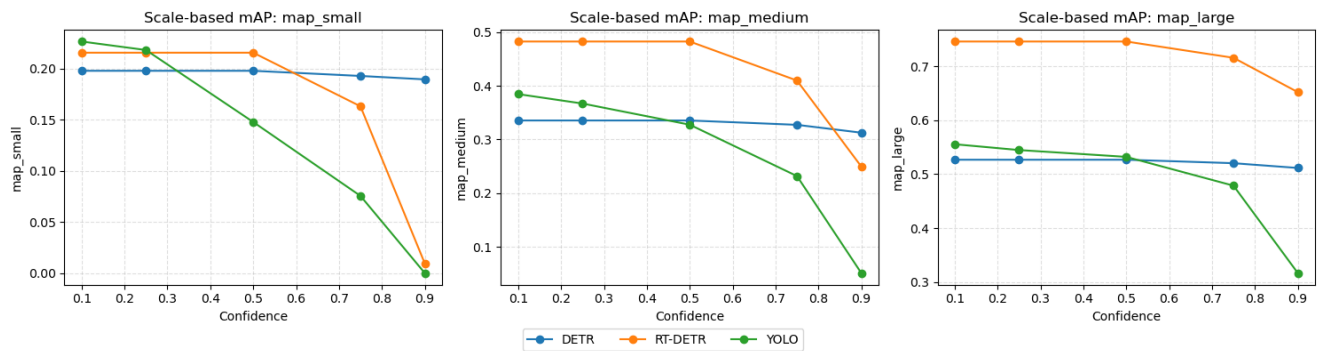


Figure 3. Task 1 — Small, medium and large mAP.

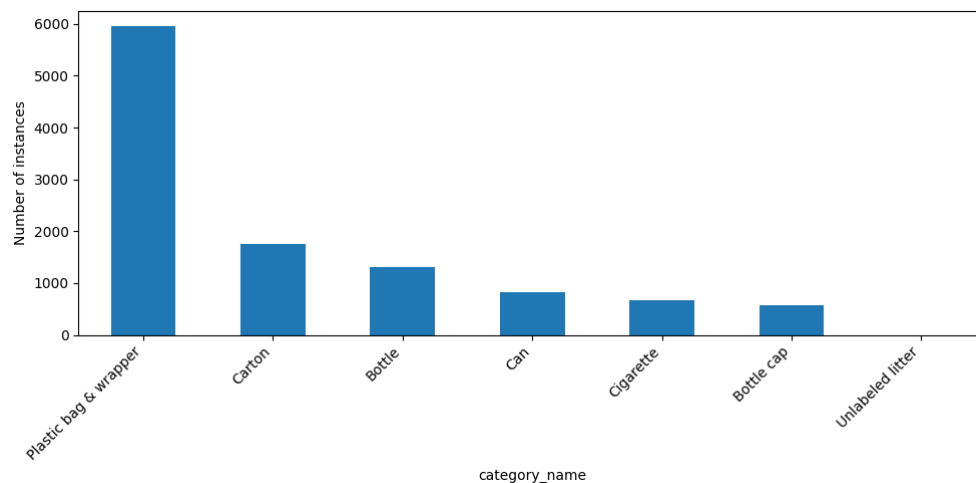


Figure 4. Task 2 — Class distribution.

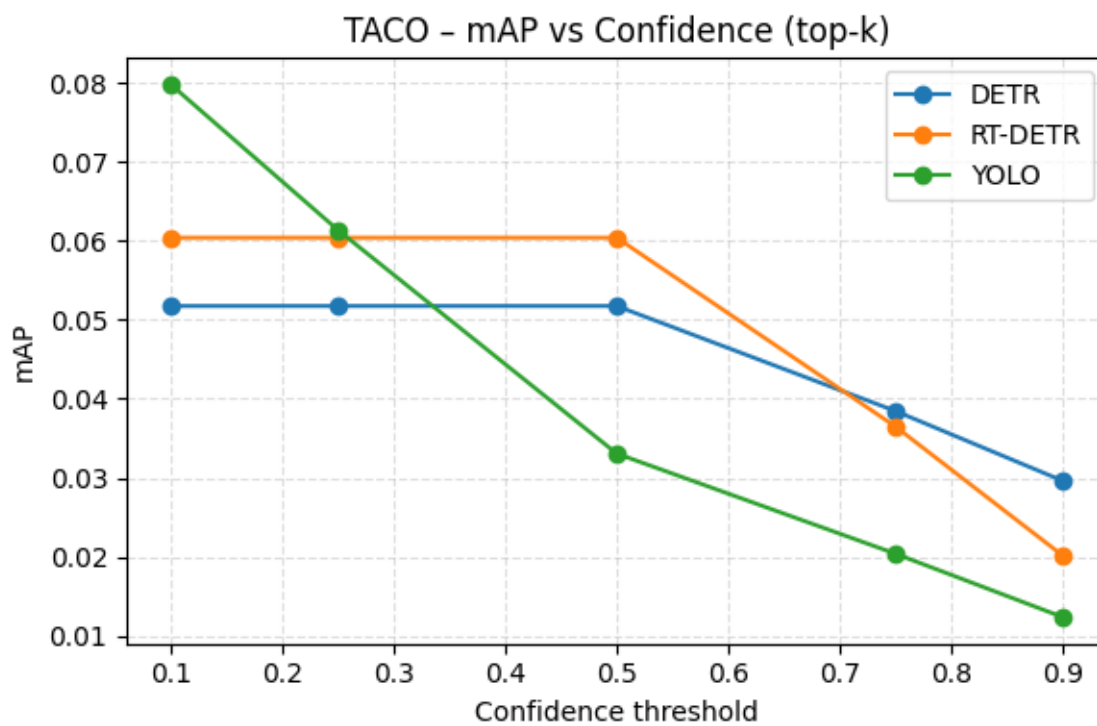


Figure 5. Task 2 — mAP vs confidence (top-k approach).

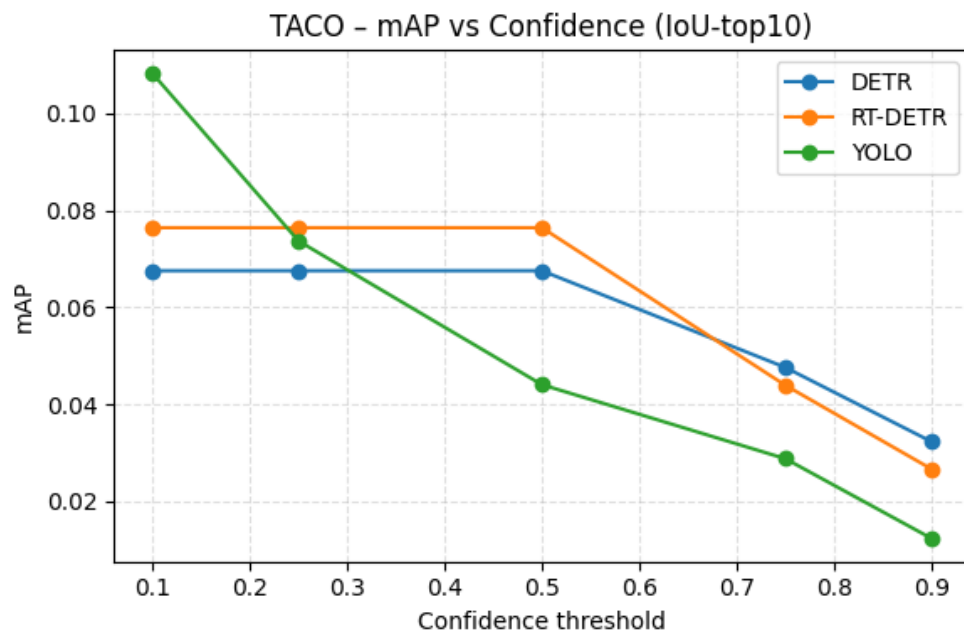


Figure 6. Task 2 — mAP vs confidence (IoU-top10 approach).

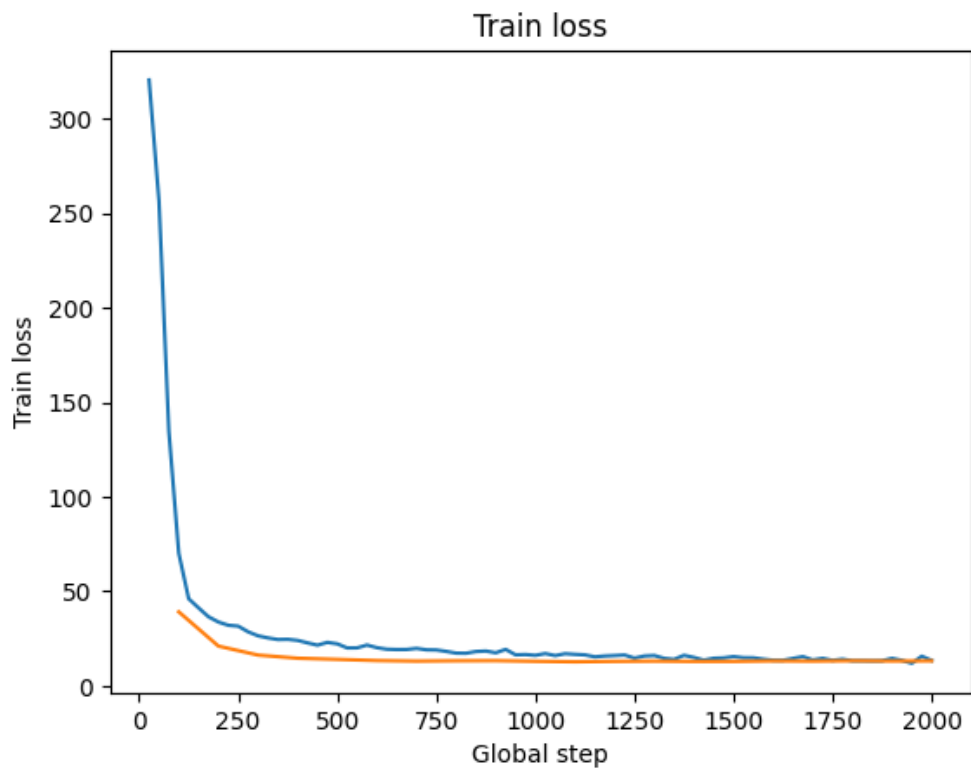


Figure 7. Task 2.3 — Training loss.

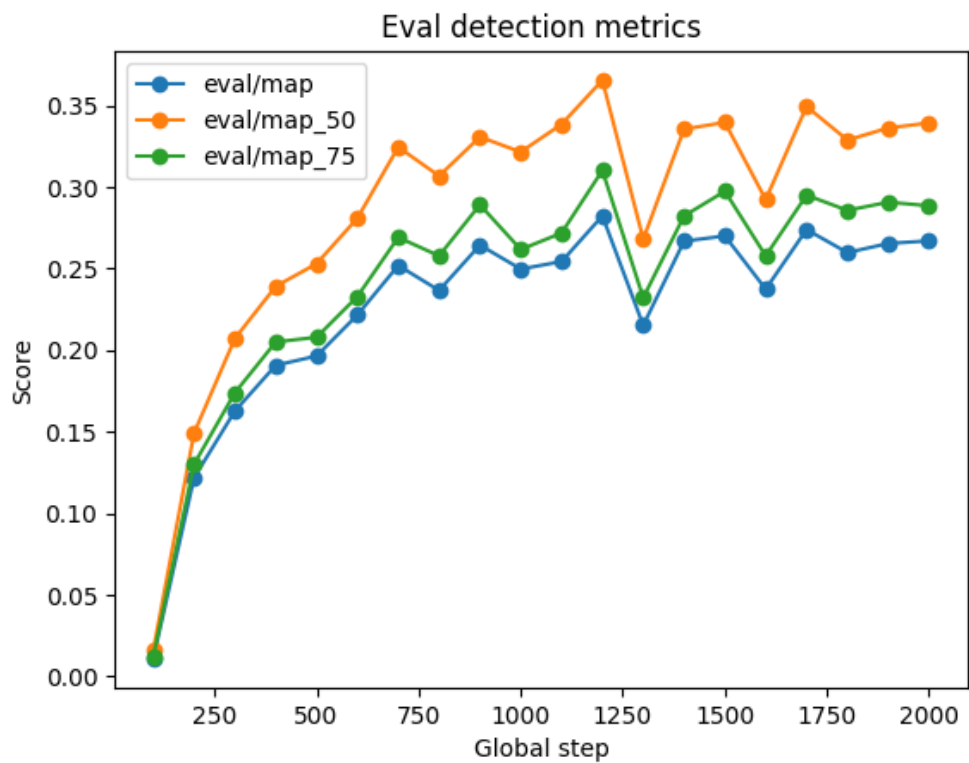


Figure 8. Task 2.3 — Evaluation metrics.

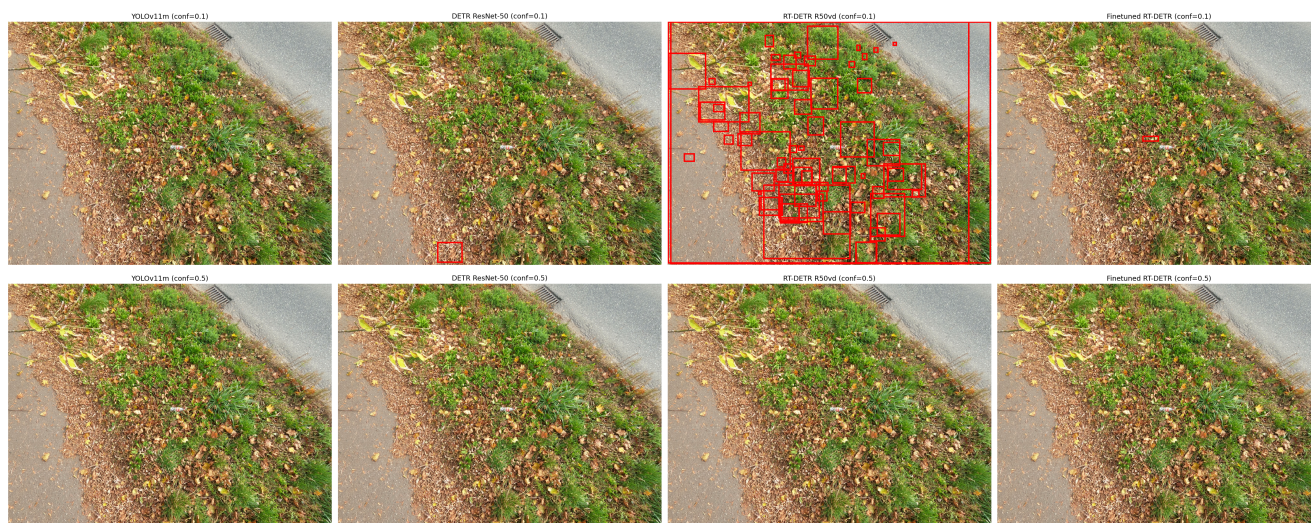


Figure 9. Task 3 — First example (0.10 vs 0.50 confidence).



Figure 10. Task 3 — Second example (0.10 vs 0.50 confidence).